Search Methods for UAV Trajectory and Packet Routing in Multi-Hop IoT-UAV Networks

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1 Overview and Four-Phase Search Strategy

Problem **P1** (defined in the companion model document) couples UAV trajectories and packet flows. A naïve search over both simultaneously is intractable. We therefore decompose the search into four sequential phases executed before the main iterative loop:

Phase 0 — Base-station strength ranking and area pre-division (§2). Rank base stations by a *strength metric* and partition the mission area into BS zones proportional to strength. This creates a coarse, BS-anchored frame of reference for all subsequent decisions and is done *once* before any drone trajectories are considered.

Phase 1 — IoT node classification (§3). Every IoT node is labelled as *proximal*, *contested*, or *isolated* based on its position relative to BS zones and UAV reachability. The label determines how much search effort is spent on that node: proximal nodes are hardcoded to the nearest BS zone; isolated nodes are hardcoded to the nearest relay chain; contested nodes receive full search.

Phase 2 — Per-UAV minimum spanning cost trees (§4). A *minimum spanning cost tree* (MSCT) is grown around each UAV's initial gateway using Prim's algorithm on the IoT graph $\mathcal{H}$. Branches are pruned whenever the accumulated multi-hop cost to a node exceeds twice the direct LoS cost to the UAV. The resulting tree defines the *effective collection neighbourhood* $\mathcal{N}_n^{\text{eff}}$ of U_n: the UAV's search need only cover gateways inside this neighbourhood.

Phase 3 — Joint iterative search (§5–§9). Gateway selection, mission division, relay matching, JIALNS path planning, min-cost packet routing, and alternating charging are executed jointly and iteratively, but now restricted to the contested nodes and the pruned gateway sets from Phases 0–2.

Figure 1 shows the overall pipeline.

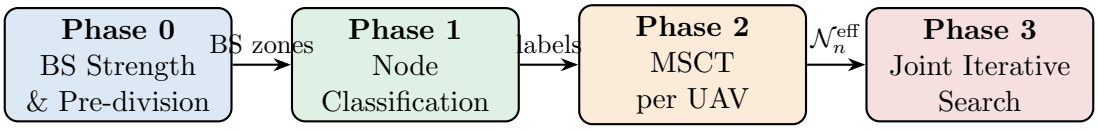

Figure 1: Four-phase search pipeline.

2 Phase 0: Base-Station Strength and Area Pre-Division

2.1 Base-Station Strength Metric

Not all base stations are equal. A station with more chargers and more assigned UAVs can service more returning drones and collect data faster. We capture this with a single scalar *strength* Ψ_l for each B_l.

Definition 1 (BS Strength)**.** *The strength of base station B_l is*

$$\Psi_l = w_c \frac{n_c^{(l)}}{n_c^{\max}} + w_n \frac{N_l^{\mathrm{UAV}}}{N}, \qquad w_c + w_n = 1, \quad w_c, w_n \geq 0, \tag{1}$$

where $n_c^{(l)}$ is the number of charging bays at B_l, $n_c^{\max} = \max_{l'} n_c^{(l')}$ normalises to the most-capable station, N_l^{UAV} is the number of UAVs initially assigned to offload at B_l, and N is the total UAV count.

Remark 1. *w_c and w_n are user-tunable. Setting $w_c = 1, w_n = 0$ prioritises charging throughput; $w_c = 0, w_n = 1$ prioritises UAV density. Equal weights $w_c = w_n = 0.5$ are the default.*

The normalised strength vector satisfies $\sum_l \Psi_l = 1$ after re-normalisation, enabling proportional area allocation.

2.2 Strength-Weighted Voronoi Pre-Division

The mission area $\mathcal{A} \subset \mathbb{R}^2$ is partitioned into L *BS zones* $\{Z_1, \ldots, Z_L\}$ using a *power-diagram* (weighted Voronoi) construction. Zone Z_l is the set of ground points closer (in the power-diagram sense) to B_l than to any other base station:

$$Z_l = \big\{\mathbf{x} \in \mathcal{A} \;\big|\; \|\mathbf{x} - \bar{\mathbf{b}}_l\|^2 - r_l^2 \;\leq\; \|\mathbf{x} - \bar{\mathbf{b}}_{l'}\|^2 - r_{l'}^2,\ \forall\, l' \neq l\big\}, \tag{2}$$

where $\bar{\mathbf{b}}_l$ is the ground projection of B_l and r_l is the *power radius*

$$r_l = r_{\min} + (r_{\max} - r_{\min})\, \Psi_l. \tag{3}$$

A stronger base station has a larger power radius, which expands its Voronoi zone proportionally. The constants $r_{\min}$ and $r_{\max}$ are set so that $|Z_l|/|\mathcal{A}| \approx \Psi_l$.

Remark 2. *The power-diagram generalises to non-convex mission areas by restricting* (2) *to the feasible region. In practice the zones are computed by rasterising $\mathcal{A}$ and assigning each cell to its nearest (power-diagram) base station.*

Each gateway g_k^* is pre-assigned to the BS zone containing it:

$$l^*(g_k^*) = \arg\min_l \Big[\|\bar{g}_k^* - \bar{\mathbf{b}}_l\|^2 - r_l^2\Big], \tag{4}$$

where $\bar{g}_k^*$ is the 2-D position of the gateway. UAVs whose subarea lies predominantly in zone Z_l are *nominated* to offload at B_l, seeding the initial relay matching in Phase 3.

3 Phase 1: IoT Node Classification

Each IoT node (or rather, each gateway g_k^* representing a component) is labelled according to three thresholds derived from the geometry of BS zones and UAV reachability.

3.1 Classification Thresholds

d_{prox}**:** proximity radius around each base station. A node within this radius of its assigned B_{l^*} has a reliable, short-range UAV relay available at all times.

d_{reach}**:** the maximum distance at which a UAV in zone Z_l can reliably collect data from a node, considering the LoS constraint (C5):

$$d_{\text{reach}} = \frac{H}{\sin(P_{\text{LoS}}^{-1}(\varepsilon))}.$$

d_{iso}**:** isolation threshold. A node beyond d_{iso} from the boundary of every BS zone cannot be reached by any UAV without an excessively long detour.

3.2 Three-Class Label Assignment

Definition 2 (Node Labels)**.** *Gateway g_k^* assigned to zone Z_{l^*} receives label*

$$\ell(g_k^*) = \begin{cases} \boldsymbol{P} \textit{ (proximal)} & \|\bar{g}_k^* - \bar{\mathbf{b}}_{l^*}\| \le d_{\text{prox}}, \\ \boldsymbol{I} \textit{ (isolated)} & \min_l \|\bar{g}_k^* - \partial Z_l\| > d_{\text{iso}}, \\ \boldsymbol{C} \textit{ (contested)} & \textit{otherwise.} \end{cases} \tag{5}$$

The three labels determine how each gateway is handled throughout the rest of the search:

Label	Colour	Meaning	Search treatment
P	green	Close to a BS	Hardcoded to nearest UAV in zone Z_{l^*}; no JIALNS search needed
C	orange	Middle / contested	Full JIALNS + min-cost flow search
I	red	Far from all zones	Hardcoded to nearest relay chain endpoint; subarea fixed, skipped in JIALNS

Hardcoding proximal and isolated nodes reduces the effective search space to the contested subset $\mathcal{G}_C^* = \{g_k^* : \ell(g_k^*) = \mathbf{C}\}$, which is typically much smaller than $|\mathcal{G}^*|$.

Proposition 1. *Let K_P, K_C, K_I denote the number of proximal, contested, and isolated gateways, respectively, with $K_P + K_C + K_I = K^*$. The search problem in Phase 3 has effective size $K_C \le K^*$, yielding a reduction factor of K_C/K^* in the JIALNS search space.*

4 Phase 2: Minimum Spanning Cost Trees (MSCT)

4.1 Motivation

Within each contested zone, a UAV does not need to visit every contested gateway. If a cluster of nearby IoT nodes can forward their data to a single gateway at a low energy cost, the UAV need only visit that one gateway. The MSCT formalises this intuition: it finds the cheapest multi-hop tree rooted at each UAV's gateway and prunes branches that are not cost-effective.

4.2 Direct LoS Reference Cost

The *direct LoS cost* of node s_i communicating with UAV U_n hovering at position $\mathbf{q}_n$ is the energy to transmit one packet directly without any relay:

$$c_{i,n}^{\mathrm{LoS}} = e_i^T \left(k_{\mathrm{pkt}},\ \sqrt{\|\bar{\mathbf{q}}_n - \mathbf{n}_i\|^2 + H_n^2} \right), \tag{6}$$

where e_i^T is the first-order radio transmission energy (Eq. (3) of the model document), $\bar{\mathbf{q}}_n$ is the horizontal projection of the UAV, and H_n its altitude.

4.3 MSCT Construction and Pruning

For UAV U_n, let $r_n \in \mathcal{G}_C^*$ be its root gateway (the contested gateway assigned to U_n by the BS zone division). The MSCT is built by a modified Prim's algorithm on the IoT graph $\mathcal{H}$ that prunes any node whose subtree path cost violates the following budget:

Definition 3 (MSCT Cost Budget)**.** *Node s_i is* eligible for inclusion *in the MSCT of U_n if and only if the total edge weight along the unique tree path from r_n to s_i satisfies*

$$c_{i,n}^{\mathrm{tree}} \triangleq \sum_{(a,b)\,\in\,\mathrm{path}(r_n,\,s_i)} w_{ab}^{II} \;\le\; 2\, c_{i,n}^{\mathrm{LoS}}. \tag{7}$$

Condition (7) ensures that routing packets through the multi-hop tree costs no more than twice the direct communication cost. The factor of 2 is an engineering bound: a larger factor admits longer, cheaper-per-hop paths but may increase AoI; a smaller factor shrinks the neighbourhood aggressively.

Remark 3. *Intra-subnet edges have $w_{ab}^{II} = 0$, so entire subnets are included "for free" whenever their gateway is within budget. Independent nodes ($\mathcal{S}^{\mathrm{ind}}$) contribute their own positive edge weights and are included only if their added cost stays within the budget.*

The effective collection neighbourhood is

$$\mathcal{N}_n^{\mathrm{eff}} = \left\{ s_i \in \mathcal{S} \;\middle|\; c_{i,n}^{\mathrm{tree}} \le 2\, c_{i,n}^{\mathrm{LoS}} \right\}. \tag{8}$$

The UAV U_n only needs to visit gateways of components that intersect $\mathcal{N}_n^{\mathrm{eff}}$.

Algorithm 1 MSCT Construction with LoS-Budget Pruning

Input: IoT graph $\mathcal{H}$, root gateway r_n, UAV position $\mathbf{q}_n$, budget factor $\kappa = 2$
Output: MSCT $\mathcal{T}_n$, neighbourhood $\mathcal{N}_n^{\text{eff}}$

```
T_n ← {r_n}; cost[r_n] ← 0; priority queue Q ← {(0, r_n)}
while Q ≠ ∅ do
    (c_cur, s_i) ← ExtractMin(Q)
    for each neighbour s_j of s_i in H do
        c_new ← c_cur + w_ij^II
        c_j^LoS ← compute via (6)
        if c_new ≤ κ · c_j^LoS and (s_j ∉ T_n or c_new < cost[s_j]) then
            cost[s_j] ← c_new; parent[s_j] ← s_i
            Add s_j to T_n; Insert(Q, (c_new, s_j))
        end if
    end for
end while
N_n^eff ← T_n
return T_n, N_n^eff
```

4.4 MSCT as a Search Bound

The MSCT provides two benefits beyond reducing the search space:

1. **Upper bound on collection energy.** Any routing solution that uses only edges in $\mathcal{T}_n$ has per-packet IoT energy at most $2 \max_i c_{i,n}^{\text{LoS}}$, guaranteeing that multi-hop never costs more than twice direct communication.

2. **Reduced TEN size.** In Phase 3, IoT-to-UAV edges in the TEN (§6) are added only for nodes $s_i \in \mathcal{N}_n^{\text{eff}}$, shrinking both $|\mathcal{N}^{\mathcal{F}}|$ and $|\mathcal{E}^{\mathcal{F}}|$.

5 Phase 3a: Gateway Selection and Mission Division

5.1 Gateway Nodes within $\mathcal{N}_n^{\text{eff}}$

Because IoT nodes can relay to one another over $\mathcal{H}$, a UAV need only visit one *gateway* per IoT component that intersects its effective neighbourhood. For each such component $\mathcal{C}_k \cap \mathcal{N}_n^{\text{eff}} \neq \emptyset$, the gateway minimises the maximum multi-hop delivery latency within $\mathcal{C}_k$:

$$g_k^* = \arg\min_{s_i \in \mathcal{C}_k} \max_{s_j \in \mathcal{C}_k} d_{\mathcal{H}}(s_i, s_j), \tag{9}$$

where $d_{\mathcal{H}}$ is the shortest-path (energy-weighted) distance on $\mathcal{H}$. Independent nodes are their own gateways. Only gateways reachable inside $\mathcal{N}_n^{\text{eff}}$ are added to U_n's visit sequence.

5.2 Strength-Weighted Mission Division

The contested gateway set $\mathcal{G}_C^*$ is divided into UAV subareas by an *energy-aware, BS-strength-weighted K-means*:

$$\text{err} = \sum_{n=1}^{N} \sum_{g \in A_n^R} \|g - \alpha_n\|^2 + \mu \max_{n,n'} |Z_n^U - Z_{n'}^U|, \tag{10}$$

where Z_n^U is the total sensor count covered by U_n. Initial centroids are seeded at the *centre of mass of each BS zone* weighted by Ψ_l, rather than randomly, to ensure that subareas respect the pre-division from Phase 0.

Algorithm 2 Strength-Seeded Mission Area Division

Input: Contested gateways $\mathcal{G}_C^*$, BS zones $\{Z_l\}$, strengths $\{\Psi_l\}$, N UAVs, load threshold γ_B, balance weight μ

Output: Subareas $\{A_1^R, \ldots, A_N^R\}$

```
Allocate N_l = ⌈N Ψ_l⌉ UAVs to zone Z_l (adjust to sum to N)
Seed N_l centroids per zone at positions of the N_l gateways in Z_l closest to b̄_l
repeat
    Assign each g ∈ G_C^* to nearest centroid
    Update centroids α_n ← (1/|A_n^R|) Σ_{g∈A_n^R} g
    Compute err via (10)
until err converges
while max_{n,n'} |Z_n^U − Z_{n'}^U| > γ_B do
    Move boundary gateway minimising flight-distance increment
end while
return {A_n^R}
```

6 Phase 3b: Packet Routing via Min-Cost Flow

6.1 Time-Expanded Network (TEN)

Given fixed UAV trajectories $\{\mathbf{q}_n[t]\}$, we build a directed time-expanded network $\mathcal{F} = (\mathcal{N}^{\mathcal{F}}, \mathcal{E}^{\mathcal{F}})$.

Node set.

$$\mathcal{N}^{\mathcal{F}} = \{(s_i, t)\}_{i,t} \cup \{(U_n, t)\}_{n,t} \cup \{(B_l, t)\}_{l,t} \cup \{v_{\text{src}}, v_{\text{snk}}\}.$$

Edge set and costs.

1. **IoT-to-IoT link** $(s_i, t) \to (s_j, t{+}1)$, active iff $(s_i, s_j) \in \mathcal{T}_n$ for some n (MSCT-restricted):

$$c_{ij}^{II} = w_{ij}^{II}, \quad u_{ij}^{II} = \min\Big(\tfrac{Q_{ij}\tau}{k_{\text{pkt}}}, B_i[t]\Big).$$

2. **IoT-to-UAV link** $(s_i, t) \to (U_n, t{+}1)$, active iff $s_i \in \mathcal{N}_n^{\text{eff}}$ and C5 holds:

$$c_{in}^{IU}[t] = \frac{e_i^T(k_{\text{pkt}}, d_{i,n}[t])}{k_{\text{pkt}}}, \quad u_{in}^{IU}[t] = \frac{Q_{i,n}[t]\,\tau}{k_{\text{pkt}}}.$$

3. **UAV-to-UAV relay** $(U_n, t) \to (U_m, t{+}1)$, active iff C6 holds:

$$c_{nm}^{UU}[t] = 0, \quad u_{nm}^{UU}[t] = \frac{Q_{n,m}[t]\,\tau}{k_{\text{pkt}}}.$$

4. **UAV-to-BS offload** $(U_n, t) \to (B_l, t{+}1)$, active iff $B_l = b_n^*$ (from relay matching):

$$c_{nl}^{UB}[t] = 0, \quad u_{nl}^{UB}[t] = \frac{Q_{n,l}[t]\,\tau}{k_{\text{pkt}}}.$$

5. **Holding edges** $(e, t) \to (e, t{+}1)$: $c_{\text{hold}} = 0$, $u_{\text{hold}}(e) = C_e$.

6. **Sink edges** $(B_l, t) \to v_{\text{snk}}$: cost $-\theta(t) = -(T - t + 1)$ (AoI freshness reward).

Proposition 2. *Maximising delivery reward $\Phi = \sum_{l,t} \theta(t) f_{nl}^{UB}[t]$ is equivalent to minimising system AoI A_P when all packet TTLs are respected. Hence P2 is consistent with the P1 AoI objective.*

6.2 Min-Cost Flow Sub-Problem (P2)

P2:

$$\min_{\{f_e \geq 0\}} \sum_{e \in \mathcal{E}^{\mathcal{F}}} c_e f_e \tag{11}$$

subject to flow balance and capacity constraints (P2-a)–(P2-b):

$$\sum_{e:\,\text{head}(e)=v} f_e - \sum_{e:\,\text{tail}(e)=v} f_e = b_v, \quad \forall v, \tag{P2-a}$$

$$0 \leq f_e \leq u_e, \quad \forall e. \tag{P2-b}$$

P2 is a classical *min-cost flow* problem solvable in polynomial time by the network simplex algorithm.

6.3 Battery-Fairness Regularisation

To bias routing toward equal battery depletion, edge costs for IoT-to-IoT and IoT-to-UAV links are augmented at each outer iteration:

$$c_{ij}^{II} \leftarrow c_{ij}^{II} + \frac{\eta_{\text{fair}}}{E_i^{\max}} \left(\frac{E_i^{\text{IoT}}}{E_i^{\max}} - \bar{\rho} \right), \tag{12}$$

where $\bar{\rho}$ is the current mean depletion and $\eta_{\text{fair}} \geq 0$ is tightened across outer iterations. This does not change the structure of P2.

7 Phase 3c: UAV Path Planning — JIALNS

7.1 Restricted Solution Representation

A candidate solution is $\mathcal{X} = (\pi_1, \ldots, \pi_N)$ where π_n is an ordered sequence of *contested* gateways in $\mathcal{N}_n^{\text{eff}}$ only. Proximal and isolated gateways are absent from π_n: they are served by fixed assignments from Phases 0–1. The base station b_n^* for U_n is initialised to the nearest B_l in its BS zone and updated after each repair step.

7.2 Cost Function

$$F_c(\mathcal{X}) = \sum_{n=1}^{N} D_n(\pi_n) - \Phi(\mathcal{X}), \tag{13}$$

where $D_n(\pi_n)$ is the Euclidean path length of U_n through its contested gateways and communication points, and $\Phi(\mathcal{X})$ is the optimal value of P2 under the induced trajectories.

7.3 Destroy Operators

(D1) Random contested remove. Remove q_R contested gateways uniformly at random from any UAV sequence.

(D2) Worst-case contested remove. Remove the q_M gateways with highest detour cost $\delta_k = d(g_{k-1}, g_k) + d(g_k, g_{k+1}) - d(g_{k-1}, g_{k+1})$.

(D3) Relay-point perturbation. Remove communication point P_{nm} from both π_n and π_m; mark pair (U_n, U_m) as unmatched. This forces the repair phase to find a new rendezvous.

7.4 Repair Operators

(R1) Greedy insertion. Reinsert each removed gateway at the position minimising F_c.

(R2) Random insertion. Reinsert each removed gateway at a uniformly random feasible position.

(R3) Relay-point reinsertion. For each unmatched pair (U_n, U_m): find the rendezvous time window, compute optimal communication point P_{nm}^* minimising combined detour; apply the speed-adjustment synchronisation (Eqs. 7–9 of the model).

(R4) BS re-assignment. After each repair, reassign each UAV to the nearest non-saturated BS in its zone via (17).

7.5 Weight Update and Acceptance

Operator pair weights are updated by:

$$\omega_{ij} \leftarrow \begin{cases} \omega_{ij}, & n_{ij} = 0, \\ (1-\alpha)\,\omega_{ij} + \alpha\,\frac{s_{ij}}{n_{ij}}, & n_{ij} > 0, \end{cases} \tag{14}$$

with scores +3 (new global best), +2 (new local best), +1 (accepted non-improving), 0 (rejected). The Metropolis acceptance criterion is:

$$P_S = \begin{cases} 1, & F_c^{(k+1)} < F_c^{(k)}, \\ \exp\!\left(-\frac{F_c^{(k+1)} - F_c^{(k)}}{\lambda_A S_A}\right), & \text{otherwise}, \end{cases} \tag{15}$$

with threshold $S_A \leftarrow \lambda_A S_A$ until $S_A < \xi_A$.

Algorithm 3 JIALNS (restricted to contested gateways)

Input: Contested subareas, relay pairs L_P, MSCT neighbourhoods $\{\mathcal{N}_n^{\text{eff}}\}$, IALNS parameters $\alpha, S_A, \lambda_A, \xi_A$

Output: Contested UAV sequences $\mathcal{X}^*$

Build initial solution by TSP-heuristic on contested gateways; insert communication points via R3; assign BS via R4
$\mathcal{X}^* \leftarrow \mathcal{X}$; evaluate F_c via P2 oracle
Initialise $\omega_{ij} \leftarrow 1/9$
while $S_A > \xi_A$ **do**
    Sample (Di, Rj) $\sim \omega$
    $\mathcal{X}' \leftarrow$ Rj(Di($\mathcal{X}$))
    Speed-adjust $\mathcal{X}'$; evaluate F_c' via P2
    if $F_c' < F_c^*$ **then** $\mathcal{X}^* \leftarrow \mathcal{X}'$; s_{ij} += 3
    end if
    if $F_c' < F_c$ **then** $\mathcal{X} \leftarrow \mathcal{X}'$; s_{ij} += 2
    else if Uniform$(0,1) < P_S$ **then** $\mathcal{X} \leftarrow \mathcal{X}'$; s_{ij} += 1
    end if
    Update ω_{ij} via (14); $S_A \leftarrow \lambda_A S_A$
end while
return $\mathcal{X}^*$

8 Phase 3d: Relay Matching and Alternating Charging

8.1 Extended Relay Matching

UAVs whose subarea lies predominantly in zone Z_l are classified as *terminal* (they offload at B_l) or *relay* (they forward data to a terminal UAV closer to some B_l).

Sender preference (multi-BS generalisation). UAV U_n prefers UAV U_m over $U_{m'}$ as relay partner if

$$\min_l \|\mathbf{q}_m - \mathbf{b}_l\| \;\leq\; \min_l \|\mathbf{q}_{m'} - \mathbf{b}_l\|. \tag{16}$$

Receiver preference. Receiver U_k prefers U_n over $U_{n'}$ if $\mathcal{E}_n \geq \mathcal{E}_{n'}$ (higher energy efficiency of U_n's subarea).

Algorithm 4 Extended Relay Matching (Multi-BS)

Input: UAVs $\mathcal{U}$, BS zones $\{Z_l\}$, strengths $\{\Psi_l\}$
Output: Relay links L_P, BS assignments $\{b_n^*\}$
1: Classify each U_n as terminal (zone boundary UAV) or relay
2: Build preference lists L_n^T via (16), L_k^R via energy efficiency
3: Run Gale-Shapley deferred acceptance on L^T, L^R
4: Assign $b_n^* \leftarrow \arg\min_l \|\mathbf{q}_n - \mathbf{b}_l\|$ for terminal UAVs
5: **return** L_P, $\{b_n^*\}$

8.2 Alternating Charging with Multiple Base Stations

Each UAV U_n charges at the nearest available B_l at each charging event:

$$b_n^{\text{charge}}(j) = \underset{B_l:\, q_l^{\text{queue}}(j) < m}{\arg\min} \|\bar{\mathbf{q}}_n(j) - \mathbf{b}_l\|. \tag{17}$$

Subareas are exchanged every T_C rounds as in the reference scheme, but UAV-to-BS assignments are updated at each exchange.

9 Joint Iterative Algorithm

10 Complexity Analysis

Let $K_C = |\mathcal{G}_C^*|$ be the number of contested gateways after classification. Table 1 summarises each component.
Here P is the number of raster cells for Voronoi construction. The critical reduction is $K_C \leq K^*$ in the JIALNS term: when most gateways are proximal or isolated (common in dense deployments near base stations), $K_C \ll K^*$ and the iterative search collapses to a small contested core.

11 Full Information-Flow Diagram

Figure 2 shows all algorithm interactions. The dashed arrows are feedback paths that only activate during Phase 3.

Table 2 contrasts the proposed scheme against the reference MGDC paper and the initial search scheme without the new phases.

Table 1: Computational complexity of each component.

Component	Algorithm	Complexity
BS strength + Voronoi	Power-diagram rasterisation	$\mathcal{O}(LP)$
Node classification	Distance thresholding	$\mathcal{O}(K^{*}L)$
MSCT per UAV	Prim's + LoS budget	$\mathcal{O}(M \log M)$ per UAV
Gateway selection	Eccentricity on $\mathcal{H}$	$\mathcal{O}(K_C^2)$
Strength-seeded K-means	Algorithm 2	$\mathcal{O}(K_C N I_{\mathrm{km}})$
Relay matching	Gale-Shapley	$\mathcal{O}(N^2)$
TEN construction (MSCT)	Restricted edge enumeration	$\mathcal{O}((\lvert\mathcal{T}\rvert + MN + N^2 + NL)\,T)$
Packet routing (P2)	Network simplex	$\mathcal{O}(\lvert\mathcal{N}^{\mathcal{F}}\rvert\lvert\mathcal{E}^{\mathcal{F}}\rvert \log \lvert\mathcal{N}^{\mathcal{F}}\rvert)$
JIALNS (total)	I_{out} steps on K_C	$\mathcal{O}(I_{\mathrm{out}}(NK_C + \mathrm{P2}))$
Charging schedule	BS assignment + queue	$\mathcal{O}(NLN_R)$
Phases 0–2	One-time setup	$\mathcal{O}(NM \log M + K^{*}L + LP)$
Phase 3 total	Outer loop	$\mathcal{O}(R[N^2 + I_{\mathrm{out}} \cdot \mathrm{P2} + NLN_R])$

Algorithm 5 Complete Four-Phase Joint Optimisation

Input: Network $(\mathcal{S}, \mathcal{U}, \mathcal{B})$, parameters, outer iterations R, tolerance ϵ
Output: Trajectories $\{G_L^*\}$, flows $\{f^*\}$, F^*

— Phase 0: BS strength and pre-division —
Compute Ψ_l for each B_l via (1)
Build power-Voronoi zones $\{Z_l\}$ via (2)
Assign gateways to zones via (4)

— Phase 1: Node classification —
Label all gateways $\ell(g_k^*) \in \{\mathbf{P}, \mathbf{C}, \mathbf{I}\}$ via (5)
Hardcode proximal ($\mathbf{P}$) to nearest zone UAV; hardcode isolated ($\mathbf{I}$) to nearest relay endpoint

— Phase 2: MSCT per UAV —
for each UAV U_n **do**
    Run Algorithm 1 to obtain $\mathcal{T}_n$, $\mathcal{N}_n^{\text{eff}}$
end for

— Phase 3: Iterative search —
Run Algorithm 2 (strength-seeded division)
Run Algorithm 4 (relay matching)
$F^* \leftarrow +\infty$; $\eta_{\text{fair}} \leftarrow 0$
for $r = 1, \ldots, R$ **do**
    [**Traj.**] Run JIALNS (Algorithm 3) for I_{out} steps $\Rightarrow \{G_L^{(r)}\}$
    [**Route**] Build MSCT-restricted TEN; apply (12); solve P2 $\Rightarrow \{f^{(r)}\}$, $\Phi^{(r)}$
    [**Battery**] Update $\rho_i^{(r)}$, $\Delta_{\text{batt}}^{(r)}$; $\eta_{\text{fair}} \mathrel{+}= \delta_\eta \Delta_{\text{batt}}^{(r)}$
    $F^{(r)} \leftarrow \alpha A_P^{(r)} + \beta E_{\text{total}}^{(r)} + \gamma \Delta_{\text{batt}}^{(r)}$
    if $F^{(r)} < F^*$ **then** $F^* \leftarrow F^{(r)}$; $\{G_L^*\} \leftarrow \{G_L^{(r)}\}$; $\{f^*\} \leftarrow \{f^{(r)}\}$
    end if
    if $|F^{(r)} - F^{(r-1)}| / |F^{(r-1)}| < \epsilon$ **then break**
    end if
    if $\max_i(\rho_i^{(r)} - \bar{\rho}^{(r)}) > \delta_{\text{gw}}$ **then** Re-run Algorithm 2 and Algorithm 4
    end if
end for
return $\{G_L^*\}$, $\{f^*\}$, F^*

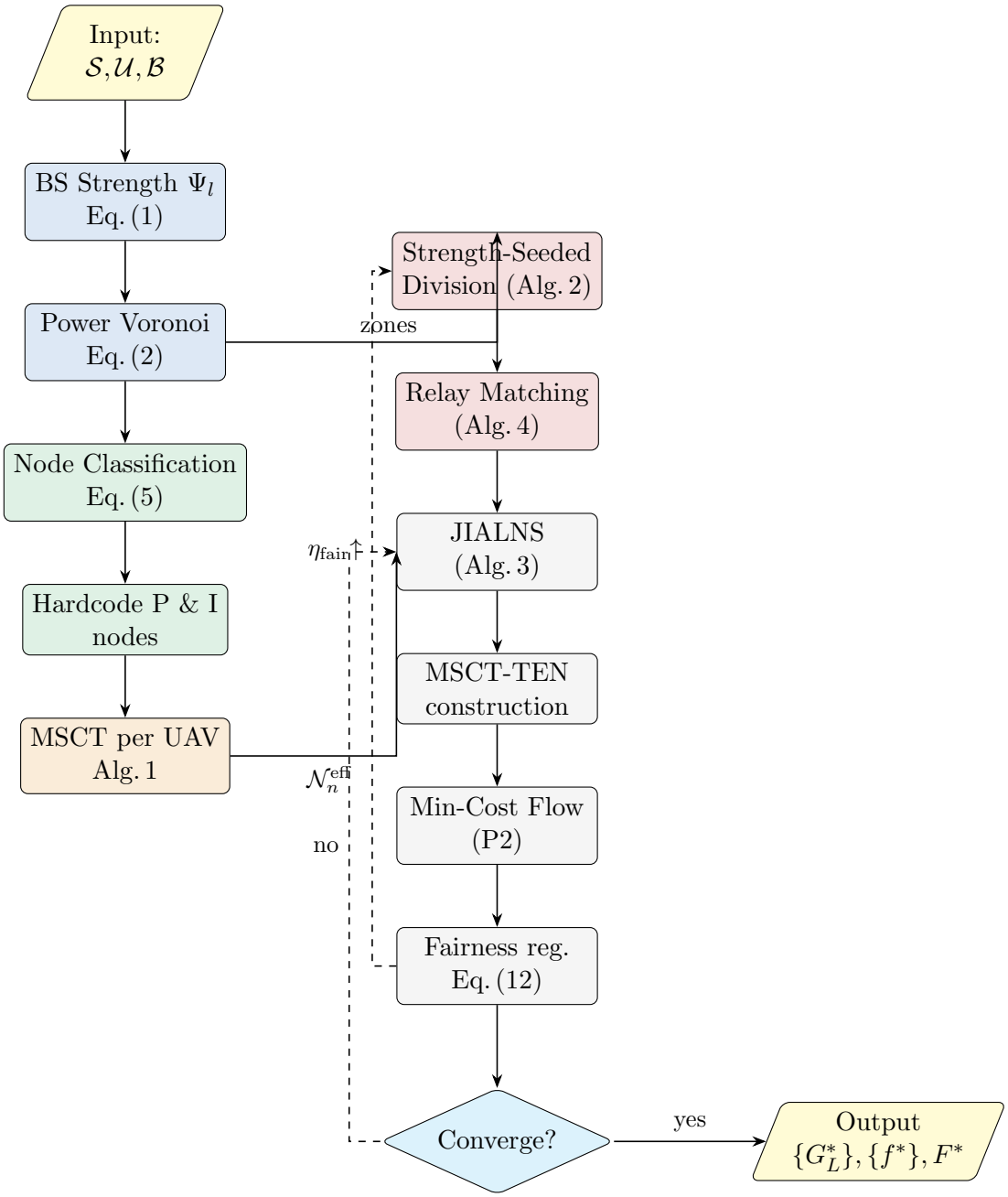

Figure 2: Full information-flow diagram of the four-phase search scheme.

Table 2: Comparison of search schemes.

Aspect	**Reference (MGDC)**	**Prior proposal**	**This work**
Area pre-division	Single BS; K-means	Multi-BS; K-means	Strength-weighted power Voronoi
Node treatment	All nodes equal	All gateways searched	P/C/I classification; hardcode P & I
Search scope	All subnets	All contested gateways K^*	Contested only $K_C \leq K^*$
Collection neighbourhood	None (visit every node)	Implicit LoS range	Explicit MSCT with 2× LoS budget
Battery fairness	Not considered	Reg. in P2 only	Reg. in P2 + strength seeding
Multi-BS charging	Not applicable	Nearest BS	Nearest non-saturated BS per zone